

Analyzing Cross Functional Roles Using a Position to Position Temporal Network

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Dataset: LinkedIn Connections Export (Personal)

Abstract

This paper investigates whether job titles co occurring within the same companies form a structured, nonrandom professional network. Using a personal LinkedIn connections export as the data source, a bipartite projection technique was applied to construct a Position to Position co-occurrence network where nodes represent job titles and edges represent shared organizational context. After cleaning, the dataset retained 1,736 valid connections drawn from 1,301 unique companies and 1,122 unique job title variants. The final projected network contained 393 nodes and 3,302 edges with a density of 0.0429. Core analysis revealed a giant component of 225 nodes, an average shortest path length of 2.61 hops, a global clustering coefficient of 0.9028, and a strong positive degree assortativity of 0.6873 confirming hub roles preferentially connect to other hub roles in a core periphery structure. The clustering coefficient of 0.9028 is approximately 22 times higher than the Erdős Rényi baseline of 0.0416, providing decisive evidence of Small World topology. The Louvain algorithm detected 68 distinct functional communities, the five largest of which align with identifiable professional domains: corporate management and analytics, general management and HR, data science and talent acquisition, academia and university administration, and senior leadership and HR strategy. A power law exponent of $\gamma = 0.75$ was estimated from the degree distribution, indicating extreme hub dominance with a very heavy tail. Temporal analysis from Q3 2022 through Q2 2026 revealed strong densification and a declining clustering coefficient trajectory consisting with cross industry career broadening.

1. Introduction

Professional networking platforms like LinkedIn generate large volumes of implicit relational data. When a person lists their employer and job title, they are unknowingly participating in a co-occurrence record: every person working at the same company contributes to that company's organizational footprint, regardless of whether they know each other personally. Aggregated across thousands of connections and hundreds of companies, these co-occurrences create a rich latent network structure that has not been widely studied at the individual network level.

This project asks a focused research question:

Do job titles co occurring within the same company form a structured, nonrandom network and if so, what does that structure tell us about how professional roles relate to each other across industries?

The motivation is both academic and practical. Academically, applying network science tools to professional role data extends the study of complex networks into a domain typically analyzed with simpler co-occurrence or keyword methods. Practically, understanding which job titles act as structural bridges connecting otherwise separate professional communities has direct

implications for career planning, organizational design, and workforce analytics. The finding that "data analyst" sits at the top of betweenness centrality in this dataset, for example, has concrete interpretive value beyond what any summary statistic could provide.

The paper is structured as follows: Section 2 describes the data and network construction methodology. Section 3 presents core network metrics. Section 4 covers centrality analysis. Section 5 presents community detection results. Section 6 analyzes the degree distribution. Section 7 compares the real network against theoretical models. Section 8 presents temporal evolution analysis. Section 9 discusses findings and limitations. Section 10 concludes with suggestions for future work.

2. Data and Methodology

2.1 Data Source

The dataset was obtained by exporting a personal LinkedIn connections list, which LinkedIn provides as a downloadable CSV file through its data export tool. The raw export contained 1,863 rows, each representing one LinkedIn connection with three key fields: the connection's name, their listed job title (Position), and their listed employer (Company). No additional private or third party data sources were used.

Data cleaning involved three steps. First, rows with missing Company or Position values were dropped. Second, all text fields were standardized to lowercase and stripped of leading and trailing whitespace, ensuring that variations like "Software Engineer" and "software engineer" were treated as the same role. Third, the "Connected On" date field was parsed into quarterly time periods using the pandas Period type, enabling temporal analysis. After cleaning, the dataset retained 1,736 valid rows drawn from 1,301 unique companies and 1,122 unique position variants. The connection dates span from Q3 2022 through Q2 2026.

2.2 Network Construction Bipartite Projection

The raw data naturally forms a bipartite graph: one partition contains Companies and the other contains Job Titles. Each person in the dataset creates an implicit edge between their employer and their job title. This bipartite structure was projected onto the job title partition to create a unipartite Position to Position network using the following rule

If two distinct job titles both appear within the same company, they receive an edge. Each additional company where the same pair co-occurs increments the edge weight by one.

This was implemented in Python using the NetworkX library. For every company with more than one distinct position, all pairwise combinations of those positions were enumerated using `itertools.combinations`, and edges were created or their weights incremented accordingly.

Isolated nodes job titles appearing only in companies with a single unique role were removed from the graph after construction.

The resulting network contained 393 nodes and 3,302 weighted edges. The average degree was $2 * 3,302 / 393 = 16.80$, meaning each job title was directly co associated with approximately 17 other job titles on average across the full dataset.

3. Core Network Analysis

3.1 Summary of Metrics

The full set of core structural metrics obtained from the network is presented below:

Metric	Value
Total Nodes (Job Titles)	393
Total Edges (Co-occurrences)	3,302
Total Connections (Raw, after cleaning)	1,736
Unique Companies	1,301
Network Density	0.0429
Size of Giant Component	225 nodes
Avg Shortest Path Length (Giant)	2.6100
Global Clustering Coefficient	0.9028
Average Clustering Coefficient	0.6867
Degree Assortativity	0.6873

3.2 Network Density

Network density is calculated as $2L / N(N-1) = 6,604 / 154,056 = 0.0429$. Only 4.3% of all possible job title pairs are connected, out of a theoretical maximum of 77,028 possible edges. This sparsity is entirely expected and reflects the real world specialization of organizations. A neurosurgeon and a logistics coordinator are unlikely to share an employer; the absence of their connection is meaningful, not a data gap.

3.3 Giant Component

The giant component contains 225 nodes, representing approximately 57% of all nodes in the network. The remaining nodes exist in smaller disconnected subcomponents or were removed as isolates. The 225 node giant components encompasses the roles with the broadest cross industry presence and forms the backbone of the professional role ecosystem analyzed in subsequent sections.

3.4 Average Shortest Path Length

The average shortest path length within the giant component is 2.61 hops. This means that any two job titles in the connected core of the network can be linked through fewer than three intermediate roles on average. This is the empirical signature of the Small World effect (Watts & Strogatz, 1998) despite the network having hundreds of nodes, the professional role ecosystem is remarkably compact. In practical terms, almost any career transition however seemingly distant passes through just two or three intermediate organizational contexts.

3.5 Global Clustering Coefficient

The global clustering coefficient (transitivity) is 0.9028. This is a remarkably high value: it means that in 90.28% of cases, when role A co exists with role B and role B co exists with role C, role A also co exists with role C. Triangles are nearly ubiquitous in this network.

To appreciate the significance of this, the equivalent Erdős Rényi random graph of the same size and density produces a clustering coefficient of approximately 0.0416 equal to the network's edge probability p . The real network's clustering is therefore approximately 22 times higher than what pure chance would predict. This is strong and decisive statistical evidence that the network is not random, and that real organizational logic the systematic co employment of functionally related roles drives its structure at a very high level.

3.6 Degree Assortativity

The degree assortativity coefficient is +0.6873. This strongly positive value means that high degree hub roles preferentially connect to other high degree hub roles. The network has a pronounced core periphery structure a dense, highly interconnected core of generalist roles (manager, data analyst, human resources specialist, project manager) surrounded by more peripheral specialist roles with fewer connections. This is consistent with how real organizations are structured a company's generalist functional roles (management, HR, analytics) form a dense interconnected core, while specialist technical or clinical roles remain more peripheral.

4. Centrality Analysis Hubs and Bridges

4.1 Degree Centrality The Hubs

Degree centrality measures how many other job titles a given role is directly paired with. The top five roles by degree centrality are:

Rank	Job Title	Degree Centrality
1	manager	0.2347
2	human resources specialist	0.2117
3	senior executive	0.2092
4	project manager	0.2092
5	deputy manager	0.2092

These are the organizational generalists of the dataset. "Manager" connects to approximately 23.5% of all other job titles in the network meaning out of 392 other nodes, it is directly linked to roughly 92 of them. These roles appear across nearly every sector in the dataset (Indian corporate sector, healthcare, academia, and US based organizations) and consequently accumulate the highest number of direct co-occurrence relationships.

4.2 Betweenness Centrality the Bridges

Betweenness centrality measures how often a role lies on the shortest path between two other roles. The top five roles by betweenness centrality are:

Rank	Job Title	Betweenness Centrality
1	data analyst	0.1097
2	manager	0.0391
3	business analyst	0.0364
4	pharmacist	0.0273
5	assistant manager	0.0256

The most structurally significant finding here is data analyst, which leads betweenness centrality with a score of 0.1097 nearly three times higher than the second ranked role. Data analysts have high degree centrality (0.2015) but is not the top degree node yet it sits on the shortest path between more role pairs than any other position in the network. This makes it the dominant structural bridge in the entire dataset.

The presence of pharmacists in the top five betweenness roles is particularly revealing. With a degree centrality of only 0.1837, it is not a top hub, yet it bridges healthcare organizations with corporate and analytics organizations in a way that few other roles do. It connects professional communities that would otherwise have long path distances between them.

The scatter plot of degree versus betweenness (Figure 1 in the appendix) shows the full picture. Most roles cluster near the origin with low values on both axes. The top bridge hub roles (data analyst, manager) sit in the top right, while pharmacist occupies the high betweenness, moderate degree quadrant the position of a structurally indispensable connector.

This finding aligns with Burt's (2004) structural holes theory individuals who occupy bridge positions between otherwise disconnected groups accumulate non redundant information and experience disproportionate career advantages.

5. Community Detection

5.1 Methodology Louvain Algorithm

The Louvain algorithm was applied to partition the network into communities by maximizing modularity a score measuring whether edges fall within communities more than would be expected in a random null model. The algorithm is entirely structure driven: it has no knowledge of what the job title strings mean and operates purely on edge connections and weights.

5.2 Results

The algorithm detected 68 distinct communities. The five largest, along with their top roles by degree centrality, are:

Community	Size	Top Roles	Inferred Domain
1	69 roles	senior executive, assistant manager, business analyst, pharmacist, senior business analyst, senior data analyst	Corporate Management & Analytics
2	55 roles	manager, human resources specialist, project manager, deputy manager, data analyst, it analyst	General Management & HR

Community Size	Top Roles	Inferred Domain
3	38 roles analyst, data scientist, accounts receivable analyst, talent acquisition executive, deputy manager ai & automations, executive manager talent acquisition	Data Science & Talent Acquisition
4	27 roles assistant professor, graduate admissions advisor, its student technician, data analytics fellow, coordinator of clubs and organizations, graduate student	Academia & University Administration
5	22 roles vice president of strategic initiatives and workforce development, assistant director of human resources, director of human resources, acting president, typist	Senior Leadership & HR Strategy

5.3 Interpretation

The five communities are immediately interpretable despite the algorithm having no semantic knowledge of the role names. Community 1 reflects the Indian corporate sector's mid to senior generalist layer roles like senior executive, business analyst, and pharmacist co-occur heavily within the same companies in that ecosystem. Community 2 is general management and people operations, the roles that appear in virtually every type of organization. Community 3 captures the modern data and talent acquisition function, where data scientists and AI automation roles increasingly co-exist with HR technology roles. Community 4 is almost entirely composed of university and academic roles, reflecting the SUNY Plattsburgh connections visible in the raw data sample. Community 5 represents senior nonprofit or institutional leadership roles.

The detection of 68 total communities from a network of 393 nodes indicates a highly diverse LinkedIn network spanning multiple distinct professional ecosystems Indian corporate, US healthcare, US academic, and US nonprofit sectors all appear to be represented with sufficient density to generate their own structural communities.

6. Degree Distribution and Power Law Analysis

6.1 Observed Distribution Shape

The degree distribution histogram shows a strongly bimodal and right skewed pattern. The largest bar at the left shows that approximately 215 job titles have very low degree (1-5 connections), reflecting the large number of highly specialized or geographically specific role variants that appear infrequently across companies. There is then a secondary cluster around

degree 60-65, which corresponds to the hub roles (manager, data analyst, project manager) that appear in large numbers of companies across multiple sectors.

This bimodality rather than a smooth power law tail is a characteristic of networks built from diverse, heterogeneous datasets where two distinct populations of nodes exist: narrow specialists and broad generalists.

6.2 Power Law Test

The estimated power law exponent from linear regression on the log log scale is $\gamma = 0.75$. The log log plot shows that the red data points scatter considerably around the fitted blue dashed line, indicating that the power law fit captures the general downward trend but the distribution is not a clean textbook power law.

A γ value of 0.75, which is below the conventional real-world range of 2-3, indicates a very heavy tail with extreme hub dominance. This is consistent with the bimodal distribution: the "rich get richer" mechanism is very strong in this network. The few hub roles that appear across many industries accumulate degrees at a rate far higher than a typical scale free network. This makes physical sense given the dataset structure roles like "manager" and "data analyst" are present in virtually every company type across healthcare, corporate India, and US academia simultaneously, concentrating degree to an unusual degree.

The scatter of points around the fitted line in the log log plot, particularly at high degree values, suggests the distribution is better described as a truncated power law or a bimodal mixture distribution rather than a pure power law. This is an honest and expected finding for a finite network with heterogeneous sector composition.

7. Theoretical Model Comparison

7.1 Benchmark Models

Two synthetic benchmark graphs were constructed with identical node count ($N = 393$) and approximately matching edge counts:

Erdős Rényi (ER) model: Edges placed with probability $p = 0.0429$. Produces a Poisson degree distribution and clustering $\approx p$.

Barabási Albert (BA) model: Each new node connects to $m = 8$ existing nodes with probability proportional to degree. Produces preferential attachment and heavy tailed degree distributions.

7.2 Metric Comparison

Metric	Real Network	ER (Random)	BA (Preferential)
Avg Shortest Path Length	2.6100	2.4298	2.4033
Average Clustering Coefficient	0.6867	0.0416	0.1032
Network Density	0.0429	0.0424	0.0400

7.3 Visual Comparison

The three network visualizations using a spring layout with node size proportional to degree reveal fundamentally different structural characters.

The ER graph (green) shows nodes of roughly uniform size with edges distributed without any visible clustering or hub structure. It looks like random noise which it is. Node sizes vary only slightly because the Poisson degree distribution keeps most degrees close to the mean.

The BA graph (red) shows a densely packed core with many large hub nodes. The "rich get richer" mechanism has created a very tight, interconnected central mass with smaller peripheral nodes surrounding it. However, all nodes are clustered together rather than organized into distinct communities.

The real network (blue) shows a completely different topology: a dense central cluster of hub nodes, several satellite sub clusters of specialist roles pulled away from the main mass by long edges, and a number of small isolated peripheral clusters. This is the visual signature of a heterogeneous network with multiple professional subcommunities that are loosely connected to the central generalist hub. The separation of the satellite clusters from the main mass corresponds to the healthcare, academic, and senior leadership communities detected by Louvain.

7.4 The Small World Conclusion

The clustering comparison is the most important result in this section. The real network's average clustering of 0.6867 is

- 16.5 times higher than the ER model (0.0416)
- 6.7 times higher than the BA model (0.1032)

The path length of 2.61 is only marginally longer than ER (2.43) and BA (2.40), meaning the network retains the compact connectivity of a random graph despite having far more local structure. This combination of high clustering, short paths is the precise definition of a Small World network (Watts & Strogatz, 1998), and the magnitude of the clustering gap here leaves no ambiguity: the professional role co-occurrence network is firmly Small World.

8. Temporal Network Evolution

8.1 Methodology

Quarterly cumulative network snapshots were constructed from Q3 2022 through Q2 2026 the full span of the dataset. At each quarter Q, the network was rebuilt using only connections made on or before Q. Quarters producing fewer than 10 active nodes were excluded. Two metrics were tracked per snapshot: network size (nodes and edges) and average clustering coefficient.

8.2 Network Growth Strong Densification

The growth curves show a clear and strong densification effect: edges grew far faster than nodes throughout the observation window. Nodes (blue line) grew gradually from near zero in Q3 2022 to approximately 393 by Q2 2026 a steady, roughly linear accumulation of new job title types entering the network. Edges (orange line), by contrast, remained relatively flat from Q3 2022 through approximately Q4 2024, then surged sharply from roughly 1,400 edges in early 2025 to 3,302 edges by Q2 2026.

This late stage edge explosion with moderate new node addition is the defining pattern of network maturation. In the early phase, each new connection introduced new role types (nodes), but the network was too sparse to form many co-occurrence pairs. In the later phase, new connections increasingly came from companies already represented in the dataset, reinforcing existing co-occurrence edges rather than creating new nodes. The very steep edge growth in 2025-2026 suggests a concentrated burst of networking activity likely a new job, internship, or professional event that added many connections within already represented industries.

8.3 Clustering Coefficient Trajectory Career Broadening Signal

The clustering coefficient trajectory is the most narratively rich output of the temporal analysis. It shows:

- Q3 2022: Very low clustering (~ 0.58), reflecting an immature network with too few connections to form triangles
- Q4 2022 / Q1 2023: A sharp peak of ~ 0.81 , the highest clustering recorded in the entire observation window
- Q1 2023 through Q2 2026: A sustained and gradual decline from ~ 0.81 down to approximately 0.69 by the end of the dataset

The early peak followed by sustained decline is a clear structural signal: initial networking was tight and clustered (likely representing connections from a single institution or industry), followed by progressive broadening as connections from new professional domains were added

over time. Each time a new industry or sector was entered each new company type bringing unfamiliar role combinations the clustering coefficient fell because those new roles did not form triangles with the existing network.

This pattern is the network structural signature of career diversification: an individual whose professional connections have increasingly spanned multiple industries and functional domains rather than deepening within one. The fact that clustering stabilized around 0.69-0.76 throughout 2023-2025 before the final mild drop suggests a period of organic, multi sector networking rather than any single dramatic industry transition.

9. Discussion

9.1 What the Structure Reveals

The convergence of results across all six analyses tells a consistent story. The professional role co-occurrence network extracted from a single individual's LinkedIn connections exhibits the structural hallmarks of a complex, nonrandom network: Small World topology, a heavy tailed degree distribution, functionally interpretable community structure, strong positive assortativity, and temporal densification. None of these properties would appear in a randomly constructed network. All of them appear here.

The strong positive assortativity (0.6873) is particularly notable. Most real world social networks exhibit slightly positive assortativity, but a value approaching 0.7 suggests an unusually strong core periphery structure. This reflects the composition of the dataset: the LinkedIn connections span both a dense Indian corporate ecosystem (where roles like manager, senior executive, and business analyst cluster very tightly) and a sparser US academic and healthcare ecosystem (where specialist roles remain peripheral). The core of the network is essentially the Indian corporate generalist layer; the periphery contains the international specialist and institutional roles.

9.2 The Pharmacist Finding

One of the most interesting individual findings is the appearance of "pharmacist" as the fourth highest betweenness centrality role in the network. Pharmacist is not a business generalist it is a specific clinical credential. Its high betweenness suggests that pharmacist employing organizations in the dataset also employ a broad range of other roles that span multiple communities, making pharmacists an unexpected structural connector between the healthcare domain and the broader corporate network. This type of discovery a role whose structural importance is not obvious from its job description is precisely the value that network analysis adds over simple frequency counting.

9.3 Gamma Interpretation

The estimated γ of 0.75 requires careful interpretation. While the conventional range for textbook scale free networks is 2-3, values below 2 arise in finite networks where the degree distribution is more extreme than a pure power law would predict. In this network, the bimodal structure (large mass of low degree specialist roles plus a small group of very high degree hub roles) produces a steep log log slope. The $\gamma = 0.75$ is best interpreted not as disconfirming scale free behavior, but as indicating a more extreme version of it one where hub dominance is stronger than in typical social networks, consistent with the very high assortativity value and the dense core visible in the network visualization.

9.4 Limitations

Several limitations apply. The dataset reflects one individual's professional network and cannot be generalized without replication across diverse users. The geographic and sectoral composition heavily weighted toward Indian corporate roles plus a SUNY Plattsburgh academic community introduces a structural bias that inflates assortativity and clustering within those domains. Job title normalization was minimal; "deputy manager," "deputy manager ai & automations," and "executive manager talent acquisition" are treated as distinct nodes despite overlapping semantically. Finally, LinkedIn connections are voluntary and subject to selection bias people connect preferentially within their primary industry and with people they find professionally relevant.

10. Conclusion and Future Work

10.1 Summary of Findings

This project constructed a Position to Position co-occurrence network from 1,736 cleaned LinkedIn connections, producing a graph of 393 nodes and 3,302 edges. The key confirmed findings are:

Small World topology confirmed: Clustering coefficient of 0.9028 (global) and 0.6867 (average) against an ER baseline of 0.0416 approximately 16-22 times higher. Average path length of 2.61 comparable to random models.

Scale free / heavy tailed distribution confirmed: Strongly right skewed degree histogram, bimodal structure, estimated $\gamma = 0.75$ indicating extreme hub dominance.

68 functional communities detected: All five large communities are interpretable as real professional domains despite the algorithm having no semantic knowledge of role names.

Strong core periphery structure: Degree assortativity of +0.6873 confirms hub roles (manager, data analyst, project manager) cluster together in a dense professional core.

Data analyst is the dominant bridge role: Betweenness centrality of 0.1097 is nearly three times the second ranked role making it the most structurally important cross functional connector in the network.

Temporal densification confirmed: Edge growth far outpaced node growth in 2025-2026. Declining clustering coefficient from 0.81 to 0.69 across the 2022-2026 window reflects career diversification across multiple professional domains.

10.2 Future Directions

Scale to multiuser datasets: Collecting LinkedIn exports from dozens of users across different industries would allow testing whether the topology generalizes and whether users from different career backgrounds produce structurally different networks.

Job title normalization via NLP: Applying transformer based embeddings to canonicalize title variants ("sr. data analyst" - "data analyst") would reduce the 1,122 position variants to a cleaner node set and likely reveal even stronger community structure.

Graph embeddings for career recommendation: Node2vec or GraphSAGE embeddings trained on this network would encode structural similarity between roles, enabling a data driven career path recommender that suggests lateral moves based on network proximity rather than keyword matching.

Link prediction for hiring gap analysis: A link prediction model trained on temporal snapshots could identify role pairs that are structurally positioned to co-occur but currently unconnected to a hiring gap or cross functional opportunity signal for organizations.

Multi layer network analysis: Adding a second layer that represents person to person connections alongside the role to role layer would create a multiplex network, allowing comparison of structural properties across layers.

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